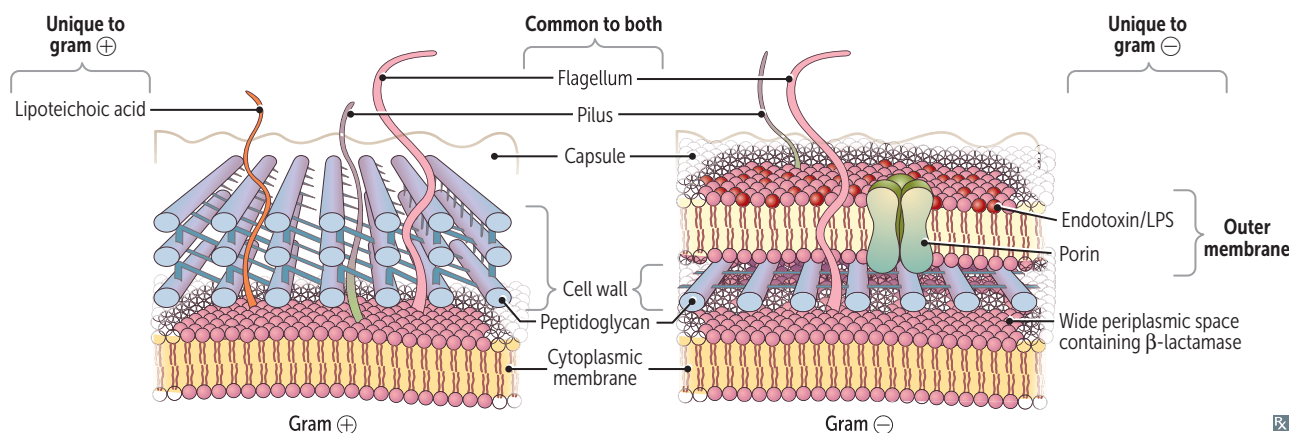


► MICROBIOLOGY—BASIC BACTERIOLOGY

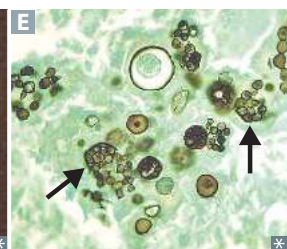
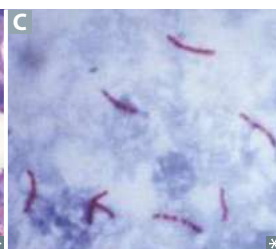
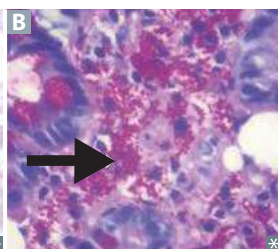
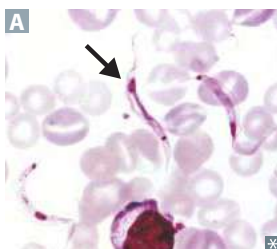
Bacterial structures

STRUCTURE	CHEMICAL COMPOSITION	FUNCTION
Appendages		
Flagellum	Proteins	Motility
Pilus/fimbria	Glycoprotein	Mediate adherence of bacteria to cell surface; sex pilus forms during conjugation
Specialized structures		
Spore	Keratinlike coat, dipicolinic acid, peptidoglycan, DNA	Survival: resist dehydration, heat, chemicals
Cell envelope		
Capsule	Discrete layer usually made of polysaccharides (and rarely proteins)	Protects against phagocytosis
Slime (S) layer	Loose network of polysaccharides	Mediates adherence to surfaces, plays a role in biofilm formation (eg, indwelling catheters)
Outer membrane	Outer leaflet: contains endotoxin (LPS/LOS) Embedded proteins: porins and other outer membrane proteins (OMPs) Inner leaflet: phospholipids	Gram $\ominus$ only Endotoxin: lipid A induces TNF and IL-1; antigenic O polysaccharide component Most OMPs are antigenic Porins: transport across outer membrane
Periplasm	Space between cytoplasmic membrane and outer membrane in gram $\ominus$ bacteria (peptidoglycan in middle)	Accumulates components exiting gram $\ominus$ cells, including hydrolytic enzymes (eg, β -lactamases)
Cell wall	Peptidoglycan is a sugar backbone with peptide side chains cross-linked by transpeptidase	Netlike structure gives rigid support, protects against osmotic pressure damage
Cytoplasmic membrane	Phospholipid bilayer sac with embedded proteins (eg, penicillin-binding proteins [PBPs]) and other enzymes Lipoteichoic acids (gram p ositive) only extend from membrane to exterior	Site of oxidative and transport enzymes; PBPs involved in cell wall synthesis Lipoteichoic acids induce TNF- α and IL-1

Cell envelope

Stains

Gram stain	First-line lab test in bacterial identification. Bacteria with thick peptidoglycan layer retain crystal violet dye (gram ⊕); bacteria with thin peptidoglycan layer turn red or pink (gram ⊖) with counterstain. These bugs do not Gram stain well (These Little Microbes May Unfortunately Lack Real Color But Are Everywhere): <i>Treponema</i>, <i>Leptospira</i> Too thin to be visualized <i>Mycobacteria</i> Cell wall has high lipid content <i>Mycoplasma</i>, <i>Ureaplasma</i> No cell wall <i>Legionella</i>, <i>Rickettsia</i>, <i>Chlamydia</i>, <i>Bartonella</i>, <i>Anaplasma</i>, <i>Ehrlichia</i> Primarily intracellular; also, <i>Chlamydia</i> lack classic peptidoglycan because of ↓ muramic acid	
Giemsa stain	<i>H. pylori</i> , <i>Chlamydia</i> , <i>Borrelia</i> , <i>Rickettsia</i> , <i>Trypanosomes</i> A , <i>Plasmodium</i>	Help! Certain Bugs Really Try my Patience
Periodic acid–Schiff stain	Stains glycogen, mucopolysaccharides; used to diagnose Whipple disease (<i>Tropheryma whipplei</i> B)	
Ziehl–Neelsen stain (carbol fuchsin)	Acid-fast bacteria (eg, <i>Mycobacteria</i> C , <i>Nocardia</i> ; stains mycolic acid in cell wall); protozoa (eg, <i>Cryptosporidium</i> oocysts)	Auramine-rhodamine stain is more often used for screening (inexpensive, more sensitive)
India ink stain	<i>Cryptococcus neoformans</i> D ; mucicarmine can also be used to stain thick polysaccharide capsule red	
Silver stain	<i>Helicobacter pylori</i> , <i>Legionella</i> , <i>Bartonella henselae</i> , and fungi (eg, <i>Coccidioides</i> E , <i>Pneumocystis jirovecii</i> , <i>Aspergillus fumigatus</i>)	HeLiCoPters Are silver
Fluorescent antibody stain	Used to identify many bacteria, viruses, <i>Pneumocystis jirovecii</i> , <i>Giardia</i> , and <i>Cryptosporidium</i>	Example is FTA-ABS for syphilis



Special culture requirements

BUG	MEDIA USED FOR ISOLATION	MEDIA CONTENTS/OTHER
<i>H influenzae</i>	Chocolate agar	Factors V (NAD ⁺) and X (hematin)
<i>N gonorrhoeae</i> , <i>N meningitidis</i>	Thayer-Martin agar	Selectively favors growth of <i>Neisseria</i> by inhibiting growth of gram ⊕ organisms with vancomycin , gram ⊖ organisms except <i>Neisseria</i> with trimethoprim and colistin , and fungi with nystatin Very typically cultures <i>Neisseria</i>
<i>B pertussis</i>	Bordet-Gengou agar (Bordet for <i>Bordetella</i>) Regan-Lowe medium	Potato extract Charcoal, blood, and antibiotic
<i>C diphtheriae</i>	Tellurite agar, Löffler medium	
<i>M tuberculosis</i>	Löwenstein-Jensen medium, Middlebrook medium, rapid automated broth cultures	
<i>M pneumoniae</i>	Eaton agar	Requires cholesterol
Lactose-fermenting enterics	MacConkey agar	Fermentation produces acid, causing colonies to turn pink
<i>E coli</i>	Eosin–methylene blue (EMB) agar	Colonies with green metallic sheen
<i>Brucella</i> , <i>Francisella</i> , <i>Legionella</i> , <i>Pasteurella</i>	Buffered charcoal yeast extract agar with cysteine and iron	The Ella siblings, Bruce , Francis , a legionnaire , and a pasteur (pastor), built the Sistine (cysteine) chapel out of charcoal and iron
Fungi	Sabouraud agar	“ Sab ’s a fun guy!”

Anaerobes

Examples include *Clostridioides*, *Clostridium*, *Bacteroides*, *Fusobacterium*, and *Actinomyces israelii*. They lack catalase and/or superoxide dismutase and are thus susceptible to oxidative damage. Generally foul smelling (short-chain fatty acids), are difficult to culture, and produce gas in tissue (CO₂ and H₂).

Anaerobes **Can't Breathe Fresh Air**.

Anaerobes are normal microbiota in GI tract, typically pathogenic elsewhere (eg, causes aspiration pneumonia). AminO₂glycosides are ineffective against anaerobes because these antibiotics require O₂ to enter into bacterial cell.

Facultative anaerobes

May use O₂ as a terminal electron acceptor to generate ATP, but can also use fermentation and other O₂-independent pathways.

Streptococci, staphylococci, and enteric gram ⊖ bacteria.

Intracellular bacteria**Obligate intracellular**

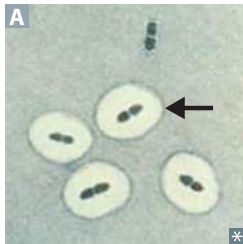
Rickettsia, *Chlamydia*, *Coxiella*
Rely on host ATP

Stay inside (cells) when it is **Really Chilly** and **Cold**

Facultative intracellular

Salmonella, *Neisseria*, *Brucella*, *Mycobacterium*, *Listeria*, *Francisella*, *Legionella*, *Yersinia pestis*

Some **Nasty Bugs May Live FacultativeLY**

Encapsulated bacteria

Examples are *Pseudomonas aeruginosa*, *Streptococcus pneumoniae* **A**, *Haemophilus influenzae* type b, *Neisseria meningitidis*, *Escherichia coli*, *Salmonella*, *Klebsiella pneumoniae*, and group B **Strep**. Their capsules serve as an antiphagocytic virulence factor.

Capsular polysaccharide +/- protein conjugate can serve as an antigen in vaccines. A polysaccharide antigen alone cannot be presented to T cells; immunogenicity can be enhanced by conjugating capsule antigens to a carrier protein.

Please **SHiNE** my **SKiS**.

Are opsonized, and then cleared by spleen.

Asplenic (**No Spleen Here**) have ↓ opsonizing ability and thus ↑ risk for severe infections; need vaccines to protect against:

- **N** meningitidis
- **S** pneumoniae
- **H** influenzae

Urease-positive organisms

Proteus, *Cryptococcus*, *H pylori*, *Ureaplasma*, *Nocardia*, *Klebsiella*, *S epidermidis*, *S saprophyticus*. Urease hydrolyzes urea to release ammonia and CO₂ → ↑ pH. Predisposes to struvite (magnesium ammonium phosphate) stones, particularly *Proteus*.

Pee **CHUNKSS**.

Catalase-positive organisms



Catalase degrades H_2O_2 into H_2O and bubbles of O_2 **A** before it can be converted to microbicidal products by the enzyme myeloperoxidase. People with chronic granulomatous disease (NADPH oxidase deficiency) have recurrent infections with certain catalase $\oplus$ organisms. Catalase $\oplus$ organisms include *Candida*, *Pseudomonas*, *Nocardia*, *Bordetella pertussis*, *Burkholderia cepacia*, *Helicobacter pylori*, *Aspergillus*, *Staphylococci*, *Serratia*, *Listeria*, *E. coli*. Catalase Positive: Notoriously Big Bubbling **HASSLE**.

Pigment-producing bacteria

<i>Actinomyces israelii</i> —yellow “sulfur” granules, which are composed of filaments of bacteria	Israel has yellow sand
<i>S aureus</i> —golden yellow pigment	<i>Aureus</i> (Latin) = gold
<i>P aeruginosa</i> —blue-green pigment (pyocyanin and pyoverdin)	<i>Aerugula</i> is green
<i>Serratia marcescens</i> —red pigment	Think red Sriracha hot sauce

In vivo biofilm-producing bacteria

<i>S epidermidis</i>	Catheter and prosthetic device infections
Viridans streptococci (<i>S mutans</i> , <i>S sanguinis</i>)	Dental plaques, infective endocarditis
<i>P aeruginosa</i>	Respiratory tree colonization in patients with cystic fibrosis, ventilator-associated pneumonia Contact lens–associated keratitis
Nontypeable (unencapsulated) <i>H influenzae</i>	Otitis media

Spore-forming bacteria

Some gram $\oplus$ bacteria can form spores when nutrients are limited. Spores lack metabolic activity and are highly resistant to heat and chemicals. Core contains dipicolinic acid (responsible for heat resistance). Must autoclave to kill spores (as is done to surgical equipment) by steaming at 121°C for 15 minutes at a pressure of 15 psi. Hydrogen peroxide and iodine-based agents are also sporicidal.

Examples: *B anthracis* (anthrax), *B cereus* (food poisoning), *Clostridium botulinum* (botulism), *Clostridioides difficile* (pseudomembranous colitis), *Clostridium perfringens* (gas gangrene), *Clostridium tetani* (tetanus).

Autoclave to kill **B**acillus and **C**lostridium (**ABC**).

Bacterial virulence factors

These promote evasion of host immune response.

Capsular polysaccharide

Highly charged, hydrophilic structure. Acts as barrier to phagocytosis and complement-mediated lysis. Major determinant of virulence.

Protein A

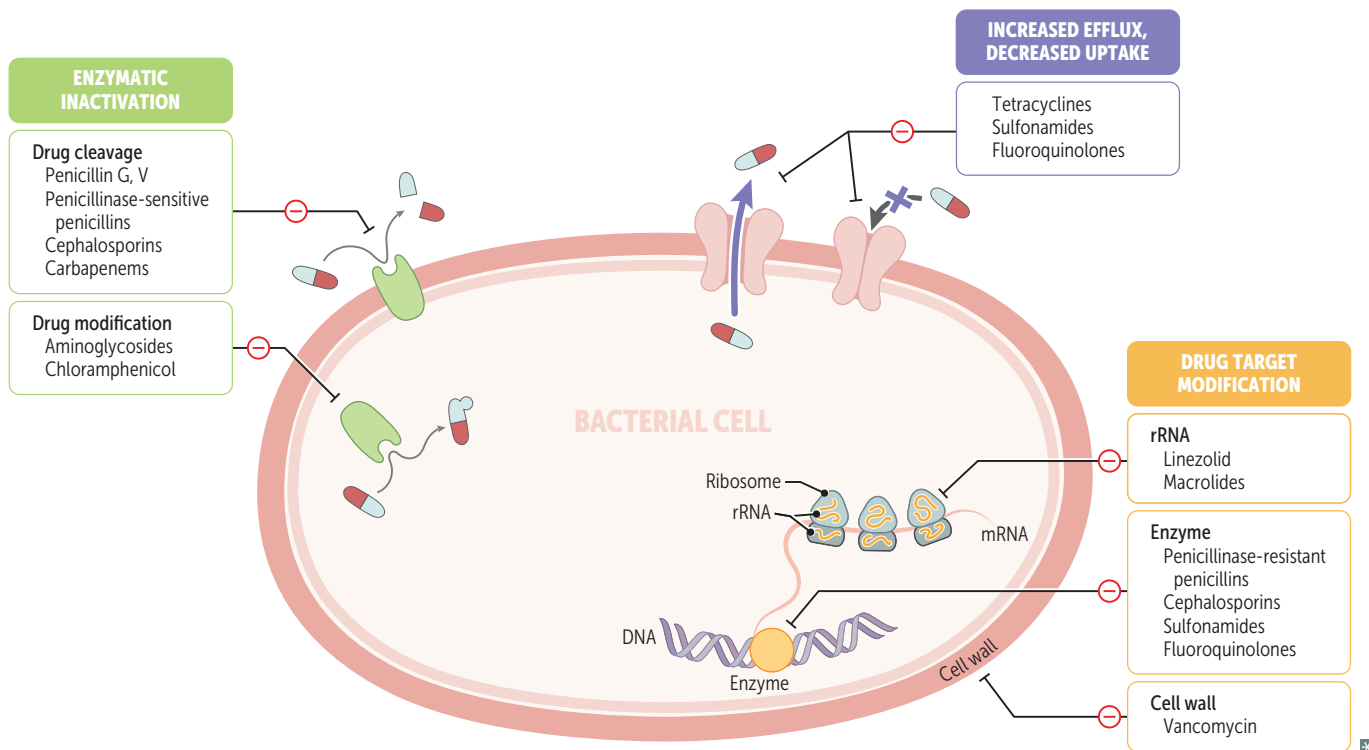
Binds Fc region of IgG. Prevents opsonization and phagocytosis. Expressed by *S aureus*.

IgA protease

Enzyme that cleaves IgA, allowing bacteria to adhere to and colonize mucous membranes. Secreted by *S pneumoniae*, *H influenzae* type b, and *Neisseria* (**SHiN**).

M protein

Helps prevent phagocytosis. Expressed by group A streptococci. Sequence homology with human cardiac myosin (**m**olecular **m**imicry); possibly underlies the autoimmune response seen in acute rheumatic fever.

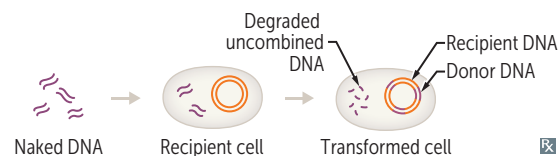
Antibiotic resistance mechanisms

Bacterial genetics**Transformation**

Horizontal gene transfer is the main mechanism for transfer of antibiotic resistance among bacteria.

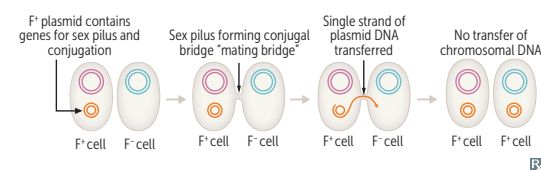
Competent bacteria can bind and import short pieces of environmental naked bacterial chromosomal DNA (from bacterial cell lysis). The transfer and expression of newly transferred genes is called transformation. A feature of many bacteria, especially *S pneumoniae*, *H influenzae* type b, and *Neisseria* (SHiN).

Adding deoxyribonuclease degrades naked DNA, preventing transformation.

**Conjugation**

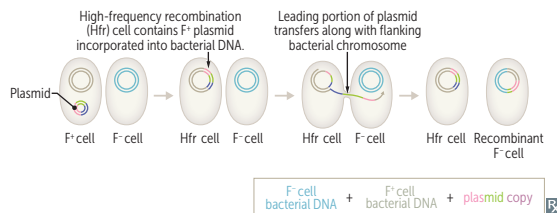
$F^+ \times F^-$

F^+ (fertility factor) plasmid contains genes required for sex pilus and conjugation. Bacteria without this plasmid are termed F^- . Sex pilus on F^+ bacterium contacts F^- bacterium. A single strand of plasmid DNA is transferred across the conjugal bridge ("mating bridge"). No transfer of chromosomal DNA.

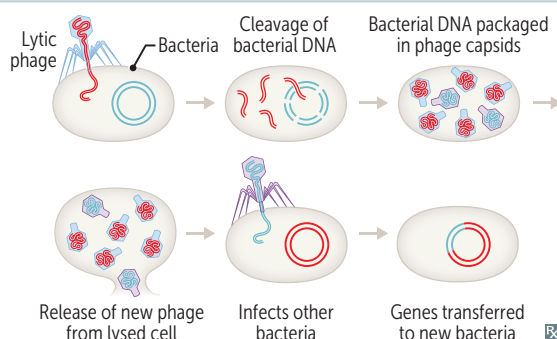


$Hfr \times F^-$

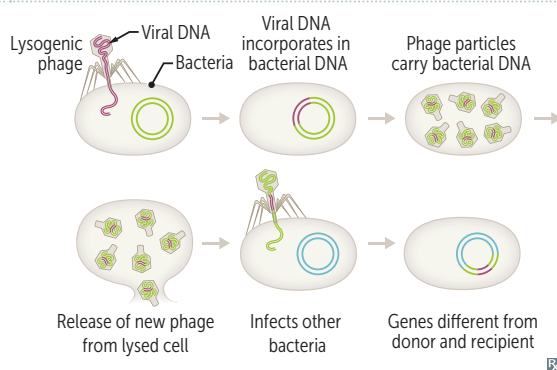
F^+ plasmid can become incorporated into bacterial chromosomal DNA, termed high-frequency recombination (Hfr) cell. Transfer of leading part of plasmid and a few flanking chromosomal genes. High-frequency recombination may integrate some of those bacterial genes. Recipient cell remains F^- but now may have new bacterial genes.

**Transduction****Generalized**

A "packaging" error. Lytic phage infects bacterium, leading to cleavage of bacterial DNA. Parts of bacterial chromosomal DNA may become packaged in phage capsid. Phage infects another bacterium, transferring these genes.

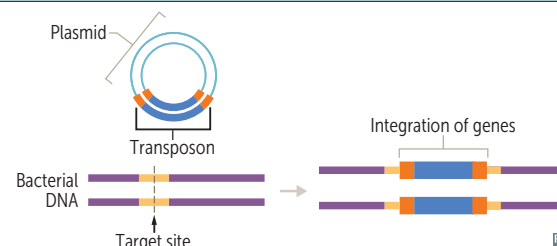
**Specialized**

An "excision" event. Lysogenic phage infects bacterium; viral DNA incorporates into bacterial chromosome. When phage DNA is excised, flanking bacterial genes may be excised with it. DNA is packaged into phage capsid and can infect another bacterium. Genes for the following 5 bacterial toxins are encoded in a lysogenic phage (ABCD'S): Group A strep erythrogenic toxin, Botulinum toxin, Cholera toxin, Diphtheria toxin, Shiga toxin.

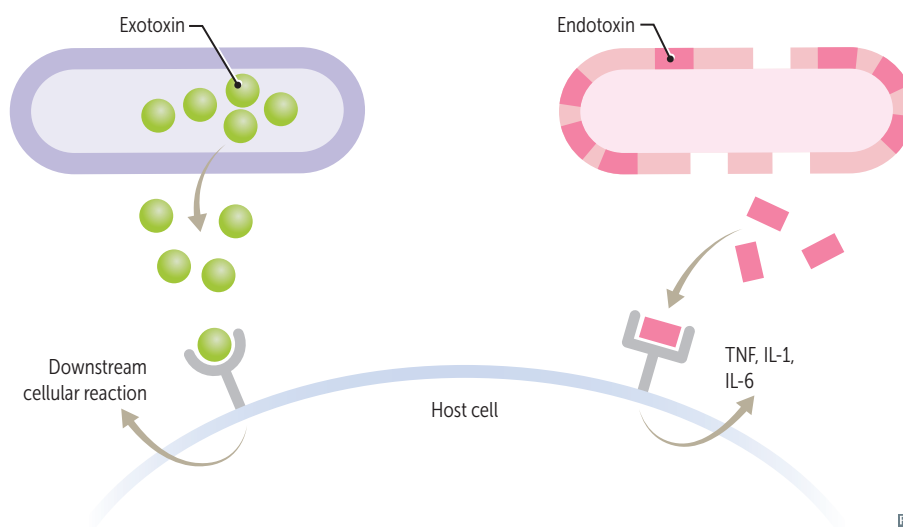


Bacterial genetics (continued)**Transposition**

A “jumping” process involving a transposon (specialized segment of DNA), which can copy and excise itself and then insert into the same DNA molecule or an unrelated DNA (eg, plasmid or chromosome). Critical in creating plasmids with multiple drug resistance and transfer across species lines (eg, Tn1546 with *vanA* from *Enterococcus* to *S aureus*).

**Main features of exotoxins and endotoxins**

	Exotoxins	Endotoxins
SOURCE	Certain species of gram $\oplus$ and gram $\ominus$ bacteria	Outer cell membrane of most gram $\ominus$ bacteria
SECRETED FROM CELL	Yes	No
CHEMISTRY	Polypeptide	Lipid A component of LPS (structural part of bacteria; released when lysed)
LOCATION OF GENES	Plasmid or bacteriophage	Bacterial chromosome
TOXICITY	High (fatal dose on the order of 1 μ g)	Low (fatal dose on the order of hundreds of micrograms)
CLINICAL EFFECTS	Various effects (see following pages)	Fever, shock (hypotension), DIC
MODE OF ACTION	Various modes (see following pages)	Induces TNF, IL-1, and IL-6
ANTIGENICITY	Induces high-titer antibodies called antitoxins	Poorly antigenic
VACCINES	Toxoids used as vaccines	No toxoids formed and no vaccine available
HEAT STABILITY	Destroyed rapidly at 60°C (except staphylococcal enterotoxin, <i>E coli</i> heat-stable toxin, and <i>B cereus</i> emetic toxin)	Stable at 100°C for 1 hr
TYPICAL DISEASES	Tetanus, botulism, diphtheria, cholera	Meningococcemia; sepsis by gram $\ominus$ rods



Bacteria with exotoxins

BACTERIA	TOXIN	MECHANISM	MANIFESTATION
Inhibit protein synthesis			
<i>Corynebacterium diphtheriae</i>	Diphtheria toxin ^a	Inactivate elongation factor (EF-2) through ADP-ribosylation	Pharyngitis with pseudomembranes in throat and severe lymphadenopathy (bull neck), myocarditis
<i>Pseudomonas aeruginosa</i>	Exotoxin A ^a		Host cell death
<i>Shigella</i> spp	Shiga toxin ^a	Inactivate 60S ribosome by removing adenine from rRNA	Damages GI mucosa → dysentery
Enterohemorrhagic <i>E coli</i>			Toxin-mediated injury and cytokine release → hemolytic-uremic syndrome (HUS; prototypically in EHEC serotype O157:H7) Unlike <i>Shigella</i> , EHEC does not invade host cells
Increase fluid secretion			
Enterotoxigenic <i>E coli</i>	Heat-labile toxin (LT) ^a	Overactivates adenylate cyclase (↑ cAMP) → ↑ Cl ⁻ secretion in gut and H ₂ O efflux	Watery diarrhea: “ labile in the Air (A denylate cyclase), stable on the G round (G uanylate cyclase)” Bacteria that ↑ cAMP include <i>V cholerae</i> , <i>B anthracis</i> , <i>B pertussis</i> , <i>E coli</i>
	Heat-stable toxin (ST)	Overactivates guanylate cyclase (↑ cGMP) → ↓ resorption of NaCl and H ₂ O in gut	
<i>Bacillus anthracis</i>	Anthrax toxin ^a	Mimics adenylate cyclase (↑ cAMP)	Likely responsible for characteristic edematous borders of black eschar in cutaneous anthrax
<i>Vibrio cholerae</i>	Cholera toxin ^a	Overactivates adenylate cyclase (↑ cAMP) by permanently activating G _s	Voluminous “rice-water” diarrhea
Inhibit phagocytic ability			
<i>Bordetella pertussis</i>	Pertussis toxin ^a	Activates adenylate cyclase (↑ cAMP) by inactivating inhibitory subunit (G _i).	Whooping cough —child coughs on expiration and “whoops” on inspiration; can cause “100-day cough” in adults; associated with posttussive emesis
Inhibit release of neurotransmitter			
<i>Clostridium tetani</i>	Tetanospasmin ^a	Both are proteases that cleave SNARE (soluble NSF attachment protein receptor), a set of proteins required for neurotransmitter release via vesicular fusion	Toxin prevents release of inhibitory (GABA and glycine) neurotransmitters from Renshaw cells in spinal cord → spastic paralysis, risus sardonicus, trismus (lockjaw), opisthotonos
<i>Clostridium botulinum</i>	Botulinum toxin ^a		Infant botulism—caused by ingestion of spores (eg, from soil, raw honey). Toxin produced in vivo Foodborne botulism—caused by ingestion of preformed toxin (eg, from canned foods)

^aAn AB toxin (also called two-component toxin [or three for anthrax]) with **B** enabling **B**inding and triggering uptake (endocytosis) of the **A**ctive **A** component. The A components are usually ADP ribosyltransferases; others have enzymatic activities as listed in chart.

Bacteria with exotoxins (continued)

BACTERIA	TOXIN	MECHANISM	MANIFESTATION
Lyse cell membranes			
<i>Clostridium perfringens</i>	Alpha toxin	Phospholipase (lecithinase) that degrades tissue and cell membranes	Degradation of phospholipids → myonecrosis (“gas gangrene”) and hemolysis (“double zone” of hemolysis on blood agar)
<i>Streptococcus pyogenes</i>	Streptolysin O	Protein that degrades cell membrane	Lyses RBCs; contributes to β -hemolysis; host antibodies against toxin (ASO) used to diagnose rheumatic fever (do not confuse with immune complexes of poststreptococcal glomerulonephritis)
Superantigens causing shock			
<i>Staphylococcus aureus</i>	Toxic shock syndrome toxin (TSST-1)	Cross-links β region of TCR to MHC class II on APCs outside of the antigen binding site → overwhelming release of IL-1, IL-2, IFN- γ , and TNF- α → shock	Toxic shock syndrome: fever, rash, shock; other toxins cause scalded skin syndrome (exfoliative toxin) and food poisoning (heat-stable enterotoxin)
<i>Streptococcus pyogenes</i>	Erythrogenic exotoxin A		Toxic shock–like syndrome: fever, rash, shock; scarlet fever

Endotoxin

LPS found in outer membrane of gram $\ominus$ bacteria (both cocci and rods). Composed of O-antigen + core polysaccharide + lipid A (the toxic component). *Neisseria* have lipooligosaccharide. Released upon cell lysis or by living cells by blebs detaching from outer surface membrane (vs exotoxin, which is actively secreted). Three main effects: macrophage activation (TLR4/CD14), complement activation, and tissue factor activation.

ENDOTOXINS:

Edema
Nitric oxide
DIC/Death
Outer membrane
TNF- α
O-antigen + core polysaccharide + lipid A
eXtremely heat stable
IL-1 and IL-6
Neutrophil chemotaxis
Shock

